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11 min
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Submitted
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Learning Check - Course of Dimensionality

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≡ Instructions

Review Learning Objectives

1. For Questions 1-4, consider the p -dimensional hypercube $H_1^p = [-1, 1]^p$. Assume we draw two p -dimensional points $\mathbf{a} = (a_1, \dots, a_p)$ and $\mathbf{b} = (b_1, \dots, b_p)$ independently and uniformly at random from H_1^p here to help.

1 / 1 point ^

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Denote by H_r^p the p -dimensional hypercube $[-r, r]^p$ for $r < 1$. What is the probability of the event $\mathbf{a} \notin H_r^p$?

Assignment details

Due

Sep 25, 11:59 PM IST

4 left (5 attempts every 24 hours)

Attempts

4 left (5 attempts every 24 hours)

Hint: Notice that, if you choose a point uniformly at random from a set $S \in \mathbb{R}^p$, the probability that this point is in another set $T \subset S$ is equal to the ratio of the volumes, i.e., $\frac{\text{Vol}(T)}{\text{Vol}(S)}$.

Submitted

Sep 23, 2:37 PM IST

Your grade

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100%

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☐ r^p

☐ πr^p

☒ $1 - r^p$

☐ $r^{(p-1)}$

Correct

Using the hint, we have that $P(\mathbf{a} \in H_r^p) = \frac{\text{Vol}(H_r^p)}{\text{Vol}(H_1^p)} = \frac{(2r)^p}{2^p} = r^p$.

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2. What is the value of r for which $P(\mathbf{a} \notin H_r^p) = 90\%$ as a function of p ?

1 / 1 point

☒ $r = 10^{(-1/p)}$

☐ $r = 90^{(-1/p)}$

☐ $r = 0.1$

☐ $r = 0.9$

Correct

Solve for $1 - r^p = 0.9$, we obtain $r^p = 0.1 = 10^{-1}$; hence, $r = 10^{-\frac{1}{p}}$

3. Using the expression for r obtained in Question 2, what is the limit of r when $p \rightarrow \infty$? Please think about the interpretation of this result.

1 / 1 point

☐ 0

☒ 1

☐ 0.5

☐ $\frac{1}{p}$

Correct

We have that $\lim_{p \rightarrow \infty} 10^{-1/p} = 1$. In other words, 90% of the mass is concentrated in a very thin layer between H_r^p and H_1^p with $r \rightarrow 1$. This result is one interesting phenomenon in high-dimensional statistics: When you pick a point uniformly at random from a particular volume, you most likely will pick a point very close to the surface of the volume when the volume lives in a high-dimensional space.

4. Compute the expected squared distance between the points $\mathbf{a}$ and $\mathbf{b}$, i.e. $E[||\mathbf{a} - \mathbf{b}||^2]$.

1 / 1 point

Hint: Remember that $||\mathbf{a} - \mathbf{b}||^2 = \sum_{i=1}^p (a_i - b_i)^2$. Also, notice that, since $\mathbf{a}$ and $\mathbf{b}$ are drawn uniformly at random from a hypercube, the coordinates a_i and b_i are independently drawn uniformly at random from the interval $[-1, 1]$; hence, their mean and variance are, respectively, 0 and $1/3$.

☐ $\frac{8}{3}$

☐ $\frac{p}{2}$

☒ $\frac{2p}{3}$

☐ p

Correct

We have that

$$E[||\mathbf{a} - \mathbf{b}||^2] = E[\sum_{i=1}^p (a_i - b_i)^2] = \sum_{i=1}^p E[(a_i - b_i)^2].$$

Since $E[(a_i - b_i)^2] = E[(a_1 - b_1)^2]$ for all values of i , we obtain

$$E[||\mathbf{a} - \mathbf{b}||^2] = \sum_{i=1}^p E[(a_1 - b_1)^2] = pE[(a_1 - b_1)^2]$$

$$= p(E[a_1^2] + E[b_1^2] - 2E[a_1]E[b_1]) = p(\frac{1}{3} + \frac{1}{3} - 0) = \frac{2p}{3},$$

where in the last line we have used that $E[a_1^2] = E[b_1^2] = 1/3$ because the points are drawn from a uniform distribution.

https://www.coursera.org/learn/machine-learning-essentials/assignment-submission/6Toflearning-check-course-of-dimensionality/view-feedback

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